(Note: This is the final report of my summer research internship at UofT)

Hand-Held Magnetic Field Generator Alignment System for a Gastrointestinal Tract Interventional Device

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1 Introduction

Endoscopic submucosal dissection (ESD) is a highly specialized endoscopic technique designed for the resection of gastrointestinal lesions in one solid piece, particularly those that are precancerous or in the early stages of cancer.

The key steps of the procedure are as follows: [1]

Insertion and Observation: The endoscope is inserted through the mouth for upper GI procedures or the anus for lower GI procedures and guided to the location of the growth.

Marking: The borders of the tumor are marked using a specialized tool passed through the endoscope.

Submucosal Injection: A solution is injected into the submucosal layer, which is the layer of tissue beneath the tumor.

Dissection: A high-frequency electrosurgical knife is used to carefully cut around and under the lesion, dissecting it from the submucosal layer.

Removal and Closure: The removed tissue is retrieved through the endoscope and sent to a laboratory for analysis.

Although ESD has many benefits, it is a technically demanding procedure which requires a high level of expertise and precision. To address these challenges, a retraction device has been developed and prototyped. This device is wireless and magnetically actuated, which allows precise control of the tissue tension. The retraction device utilizes Twisted String Actuator (TSA) mechanism. Therefore, an external magnetic handheld device has been developed to control the manipulation of the internal magnet inside retraction device [2].

However, current system is limited by the operator's ability to align the external permeant magnet with the internal permeant magnet housed within the retraction device. Firstly, the operator's view is occluded because surgical intervention occurring within the gastrointestinal tract and abdominal cavity. Secondly, even though endoscopes can provide images of the surgical workspace, these images fail to give guidance on where the hand-held device should be placed. Lastly, the alignment between internal and external magnet is not always intuitive such as "pointing to" or "in parallel" as it may requires more complex and unstraightforward math calculations. As a result, to further enhance the retraction device's usability in real ESD surgeries, an alignment system is developed to address the challenges above of current TSA retraction system. The proposed approach consists of two hall effect sensor arrays and an alignment visualization system.

2 Objectives

1. Determine the orientation of the magnetic dipole moment (m_d) of the internal permanent magnet within the magnetic retraction device, ensuring no misalignment greater than 45° about the x-axis or z-axis.
2. Guide operator towards the ideal orientation of the hand-held external actuation unit.

3 Theory

The alignment of the internal and external magnets is based on a rotating magnetic field generation approach [3]. This research focuses on developing a mathematical framework that decouples the position of the actuator magnet from the position of the internal device. This framework enables calculations for any location of the external magnet (actuator), determining the precise axis and speed at which it should rotate to generate the desired rotating magnetic field at the location of the internal magnet (device).

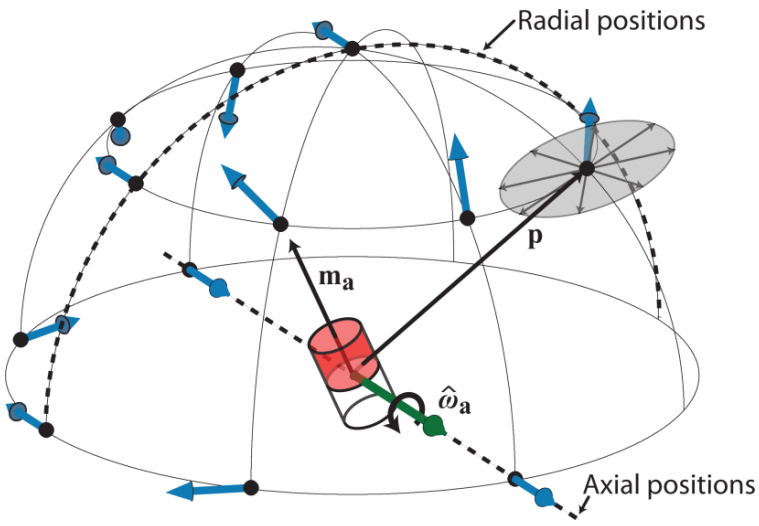


Figure 1: Notations in [3]

External magnet: (magnet at the center)

$$\text{Magnetic moment: } m_a, \textit{Rotation axis: } \widehat{w_a}$$

Internal actuation location:

$$\textit{Position vector: } p, \textit{Rotation axis: } \widehat{w_h}\ (\textit{Blue Arrows})$$

Two aspects of internal device actuation are discussed: (1) the rotation axis of the magnetic field, and (2) the magnitude and rotation speed of the magnetic field. The second aspect has already been examined during the development of the external handheld device [2].

Therefore, the primary focus of this project is the generation of the rotation axis of the magnetic field.

There are two scenarios of generating a rotating magnetic field:
$choosing\ \widehat{w_a}\ given\ \widehat{w_h}\ and\ p$ and $choosing\ \hat{p}\ given\ \widehat{w_h}\ and\ \widehat{w_a}$

Since the external magnet in the hand-held device can move freely in space, this task is in scenario 1: $choosing\ \widehat{w_a}\ given\ \widehat{w_h}\ and\ p$. The calculation procedure is shown below:

1. Find position vector p
2. Calculate transformation matrix: $H = 3\hat{p}\hat{p}^T - I$
3. Find desired rotation axis $\widehat{w_h}$
4. Calculate actuator rotation axis: $\widehat{w_a} = \widehat{H \cdot \widehat{w_h}}$

4 Problem Abstraction and Formulation

The task can be summarized as finding the correct rotation axis of the external magnet within an acceptable error tolerance and guiding the operator toward the ideal orientation. Based on the rotating magnetic field generation approach described above, this task can be formulated as follows:

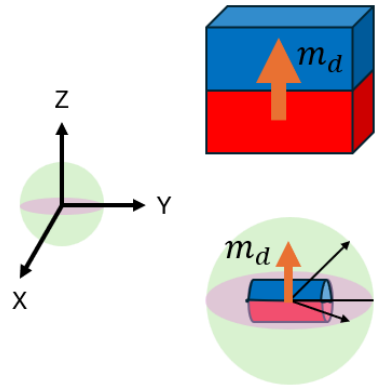


Figure 2: Internal and External Magnets [4]

4.1 Task Units

- Magnets:
    - *Internal magnet:* 5 mm long × 2.5 mm wide diametrically magnetized N50 NdFeB cylindrical magnet
    - *External magnet:* 2 in × 2 in axially magnetized N52 NdFeB cylindrical magnet
- Handheld device constraints:

- Maximum allowable misalignment: 45° about the x-axis or z-axis
- Minimum distance to the internal magnet: 3 cm (due to physical space limitations)
- Maximum distance to the internal magnet: 10 cm (due to field strength limitations)

4.2 Task Division

1. Select suitable sensors to build sensor arrays
2. Localize internal and external magnets
3. Develop working procedures and alignment methods
4. Build a real-time, visually assisted alignment system

The next section will discuss the details of these sub-tasks.

5 Methodology

5.1 Sensor Array Selection

Several parameters are critical to the overall performance of a sensor array:

- **Measurement range:** Determines the range that the sensor can measure.
- **Noise level:** Defines the accuracy and uncertainty of the measurement, often expressed as a standard deviation.
- **Resolution:** Also influences accuracy, but its effect is most significant when strict precision requirements are present.
- **Conversion time:** The time required for a single measurement. This directly impacts measurement speed and is critical for real-time processing.

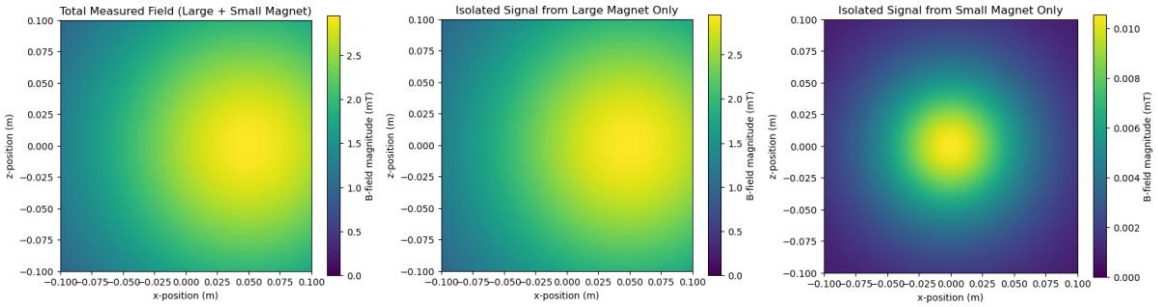
These parameters often conflict with one another. For example, a larger measurement range usually leads to higher noise levels and lower resolution, resulting in less accurate measurements. Noise can be reduced through built-in digital filters and oversampling, but this increases conversion time. Conversely, sensors with low noise levels generally have narrower measurement ranges, making them susceptible to saturation when exposed to strong magnetic fields.

To determine suitable specifications, Magpylib simulations were conducted to investigate the magnetic fields generated by the internal and external magnets.

Simulation Settings:

- Observation plane: 0.1 × 0.1 mm square in the yz-plane, centered at (0, 0, 0)
- Internal magnet: located at (0.03, 0, 0), magnetic moment along the positive x-axis
- External magnet: located at (0.15, 0.05, 0), magnetic moment along the negative x-axis

Heatmap Analysis:



Sobel Operator Analysis:



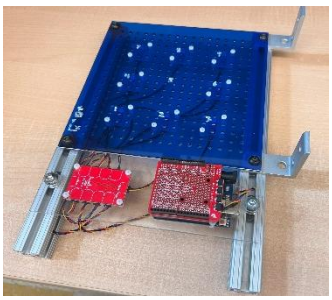
Figure 3: Heatmap Analysis and Sobel Operator Analysis for Two Magnets

Two analyses were performed: Heatmap analysis provided an overview of magnetic field strength across the region. Sobel operator analysis examined the gradient, indicating the rate of change of the magnetic field.

Both analyses showed that the external magnet dominates the total field. The external magnet generated a field on the order of a few millitesla, while the internal magnet's field was three orders of magnitude lower. Similarly, the rate of change of the magnetic field was also greater for the external magnet due to its much stronger signal.

These results highlight the large disparity in signal strength between the two magnets. The internal magnet requires sensors with high resolution and low noise to enable precise measurement, but this comes at the expense of measurement range. Conversely, the external magnet requires sensors with a wide measurement range to prevent saturation, even though this may introduce higher noise.

To resolve this trade-off, it is proposed that **two separate sensor arrays**, each equipped with different sensors, be used to localize the internal and external magnets independently. This dual-array approach eliminates the inevitable compromise between measurement range and precision.



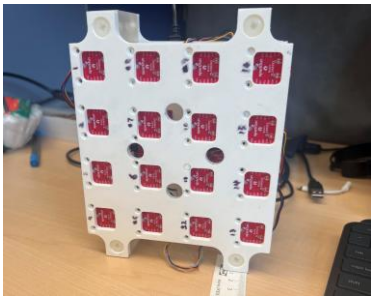


Figure 4: MMC5983 Sensor Array (left, for internal magnet, maximum range: +/-0.8mT, noise level: +/- 0.04uT*), MLX90393 Sensor Array (right, for external magnet,* maximum range: +/-50mT, noise level: +/- 5uT*)*

5.2 Magnet Localization

The localization of magnets is formulated as an optimization problem. Both the external and internal magnets are localized using the same optimization flow, but they are treated separately. The overall algorithm for localizing a single magnet (either internal or external) is as follows:

Optimizer (Trust Region Framework (TRF), cosine-distance residual, Magpylib simulation)

1. $Inputs$
   - $\mathcal{S} = \{r_1 \dots r_8\}\ sensor\ positions$
   - $B_{meas} \in \mathbb{R}^{8\times3} measured\ fields$
   - $x_0 = [\alpha_0, \beta_0, x_0, y_0, z_0]^{\mathrm{T}}\ initial\ guess$
2. $Objective$
   $Minimize\ the\ cosine\ distance\ for\ every\ sensor$:
   $$minimize\ C(x) = \sum \left\| 1 - \frac{B_{sim} \cdot B_{meas}}{\|B_{sim}\| \cdot \|B_{meas}\|} \right\|^2$$
3. $TRF\ iteration$
   $a. Place\ virtual\ cylinder\ at\ p = (x, y, z)\ with\ moment\ m$
   $b. B_{sim} \leftarrow Magnet.getB(\mathcal{S})$
   $c. Normalise\ rows \rightarrow B_{sim}\ ;\ likewise\ B_{meas}$
   $d. r(x) \leftarrow vec(B_{sim} - B_{meas})$
   $e. Trust\ region\ update\ with\ bounds$
   - $\alpha \in [-\pi, \pi]$
   - $\beta \in [0, \pi]$
   - $(x, y, z)\ unbounded$
4. $Output$
   $Estimated\ location\ and\ orientation: \dot{x} = [\dot{\alpha}, \dot{\beta}, \dot{x}, \dot{y}, \dot{z}]^{\mathrm{T}}$

Magpylib simulations are used to predict the magnetic field during optimization instead of the dipole model. Comparison tests showed that both approaches produce similar results, but Magpylib offers slightly better accuracy because it accounts for the geometry of the magnet. This is particularly important since the external magnet is relatively large and the internal magnet is often very close to the sensor array, where geometric effects potentially influence measurements. By incorporating magnet geometry, the discrepancy between real measurements and simulations is reduced, resulting in improved accuracy.

The initial guess is generated in two steps. To find the location initial guess, the sensor with the largest reading is identified, and its location is used as the initial guess for the magnet position, with a small offset applied. The initial guess of the magnetic moment orientation is computed by averaging the magnetic field vectors across all sensors, providing a rough estimate of direction.

The loss function utilizes cosine distance instead of Euclidean distance. Cosine distance is defined as

$$d(A,B) = 1 - \frac{A \cdot B}{\|A\| \cdot \|B\|}.$$

By normalizing the vectors, cosine distance removes the influence of magnitude and focuses on aligning the simulated orientation with the measured orientation for each sensor. This approach has two main advantages. Firstly, since the magnitude is canceled out, the optimization becomes invariant to the absolute magnetic strength. Euclidean distance, by contrast, is highly sensitive to errors in assumed strength. Secondly, Cosine distance treats all sensors equally, combining more directional information and reducing error. This is especially important for the internal magnet, whose field is very weak; ignoring field magnitude can help mitigate uncertainty.

The optimization is carried out using the Trust Region method. Although computationally more expensive than the traditional Levenberg–Marquardt approach, it is more robust to poor initial guesses and non-linear landscapes. Additionally, because the spherical angle system is used, boundaries can be imposed on the polar angle (constrained to $[0, \pi]$), a feature not supported by Levenberg–Marquardt.

5.2.1 Simulation Results:

To evaluate the accuracy of the localization system, Monte Carlo simulations were performed for both the internal and external magnets. The results are summarized in the graphs below. They show the mean position error and orientation error as functions of the distance along the z-axis (i.e., the distance between the magnet and the sensor array). The design goal was

to achieve position errors smaller than 5 mm and orientation errors no larger than 10°. Accordingly, the failure rate for meeting these requirements was also plotted.

Monte Carlo simulation's setting is as follows. The magnet's x coordinate and y coordinate were fixed, while the z coordinate, representing the distance from the magnet to the sensor array, was swept in a defined range. The step size is 0.1cm and at each step, 100 optimization runs were executed to evaluate the error distribution.

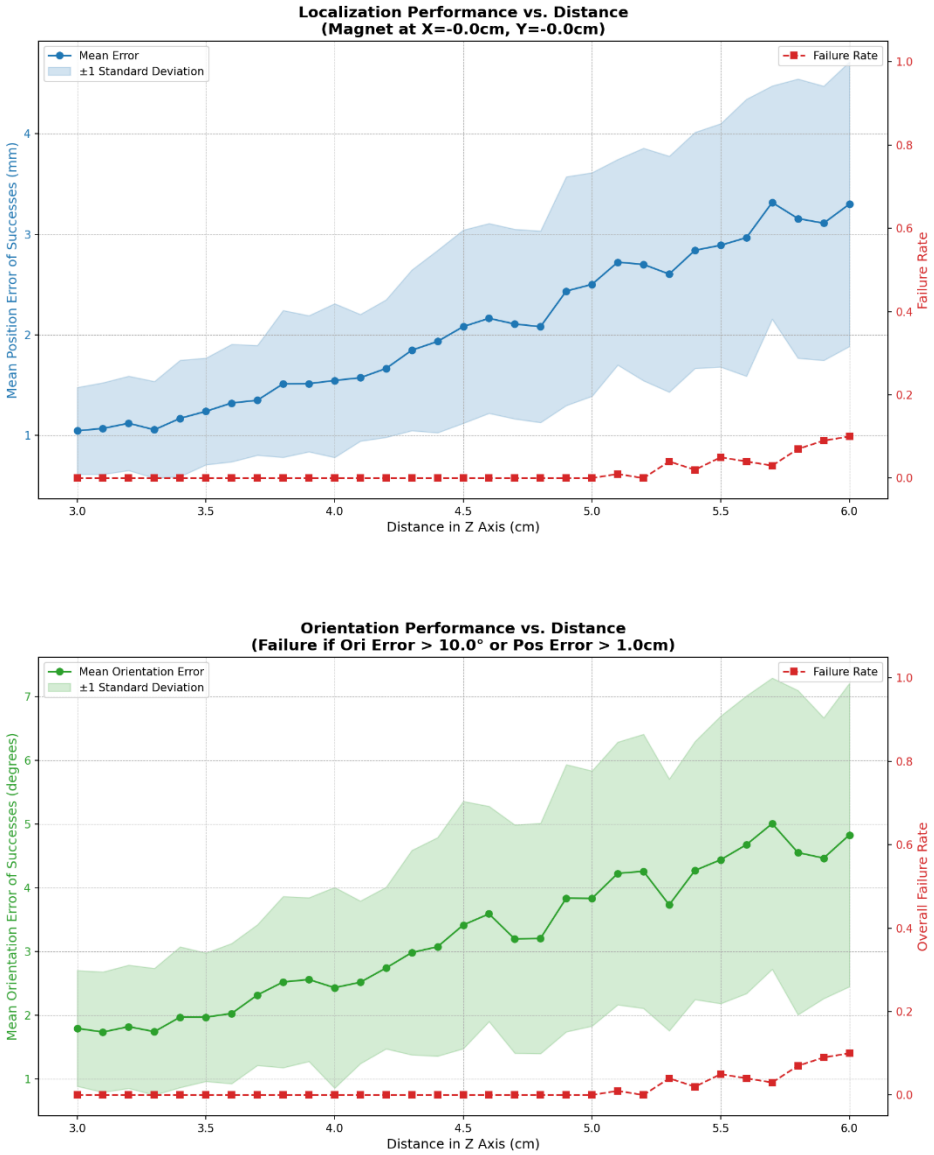


Figure 5.1: Location and Orientation Error of Internal Magnet Localization Simulation

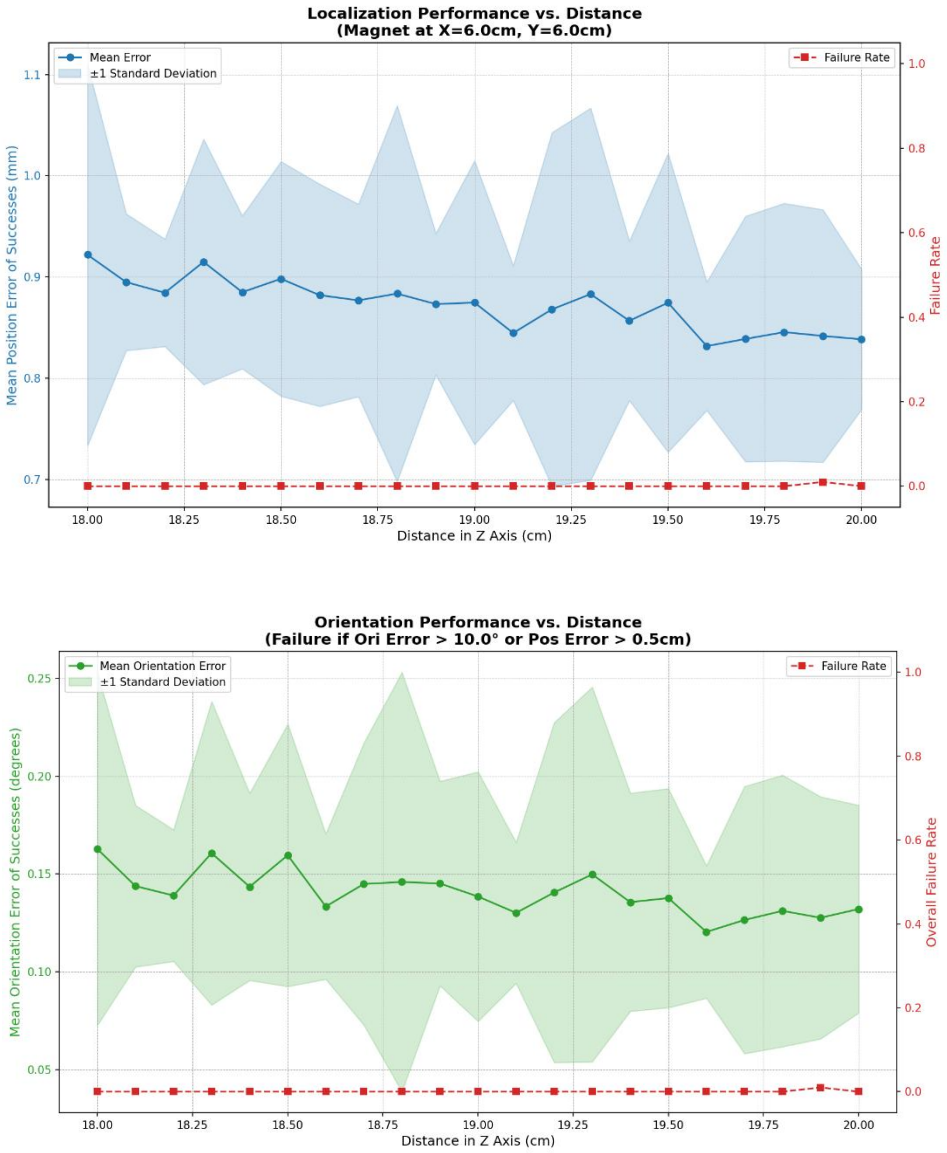


Figure 5.2: Location and Orientation Error of External Magnet Localization Simulation

Based on worst-case analysis, the localization system meets the required accuracy thresholds. For the internal magnet, the stable operation range is 3cm to 6cm from the sensor array. For the external magnet, the operation range can extend beyond 20cm from the sensor array in the worst case.

Since the distance between the external and internal magnets must remain within 10 cm, and the distance from the external magnet to the sensor array depends on sensor placement, the operation range is flexible. Nevertheless, the external magnet's 20 cm range provides sufficient tolerance to satisfy most practical scenarios.

5.3 Working Procedures and Alignment Methods

After localizing the magnets, the estimated location and orientation are fed into the rotating magnetic field generation equations to calculate the rotation axis of the external magnet. Two key challenges must be addressed.

First, this project uses two different types of sensors, each with very different measurement ranges. It is impractical to localize both magnets simultaneously because their distance is only about 10 cm. The MMC5983 sensors can easily become saturated by the external magnet and fail to detect the weaker magnetic field of the internal magnet. Therefore, the magnets are localized separately, and the internal magnet's position and orientation are assumed to be fixed once localization is completed.

Secondly, the sensor array localization can only provide an estimate of the magnetic moment direction, not the true orientation of the internal magnet. This means that one degree of freedom is missing. When the internal magnet rotates about its magnetic moment, the optimizer produces the same location and magnetic moment direction estimation. The figure below illustrates two possible situations. In *images a* and *b*, the internal magnet has different orientations but the same location and magnetic moment direction. The external magnets rotate in the direction indicated by the black arrow, resulting in identical rotating magnetic fields at the location of the internal magnet. The internal magnet in *image a* can be successfully actuated and rotates in the direction shown by the purple arrow. This purple arrow represents the effect of the rotating external magnet and is the same for both images a and b. However, in *image b*, the internal magnet cannot rotate in the direction of the purple arrow due to the constraints of the TSA device tube. In reality, the internal magnet will attempt to rotate in the direction indicated by the orange arrow but is blocked by the physical shape of the TSA device tube, preventing successful actuation.



Figure 6: Two Possible Orientations but Same Location and Magnetic Moment Direction

To solve this problem, a human-in-the-loop alignment method has been designed. It is important to note that the rotation axis has two key properties in this project: the rotation axis of the internal magnet must always align with the axis of the tube, and the desired rotation axis of the internal magnet must always be perpendicular to its magnetic moment. Since a misalignment of 45 degrees is permissible for the hand-held device, two candidate rotation axes for the internal magnet can be selected. These two axes are both perpendicular to the magnetic moment and to each other (see figure (a)). One of the axes in this pair is guaranteed to provide correct actuation. As shown in the figures below, if the magnet is oriented as in

figure (b), any rotation axis within the orange region satisfies the 45-degree requirement. Similarly, if the magnet is oriented as in figure (c), any rotation axis within the green region also meets the 45-degree requirement. Therefore, by selecting two perpendicular rotation axes, both the orange and green regions are covered, ensuring that at least one of them is a usable rotation axis.

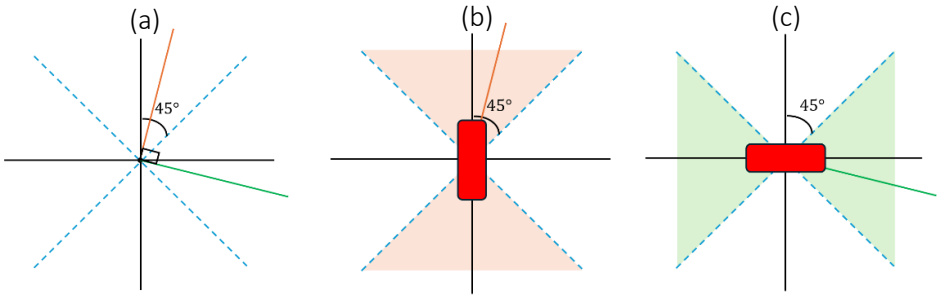


Figure 7: Alignment Methods

Here is the complete alignment procedure:

Assumptions: The internal magnet does not move after localization, misalignment does not exceed 45 degrees, and the sensor arrays are fixed in known positions.

1. The TSA retraction device is sent to the workspace through the endoscope.
2. The retraction device is secured in the workspace using surgical clips.
3. The MMC sensor array is used to localize the internal magnet within the retraction device.
4. The hand-held device is introduced, and the MLX sensor array is employed to localize the external magnet in real time.
5. Using the localization results of both the internal and external magnets, the required rotation axis of the external magnet can be calculated in real time based on the equations mentioned above.

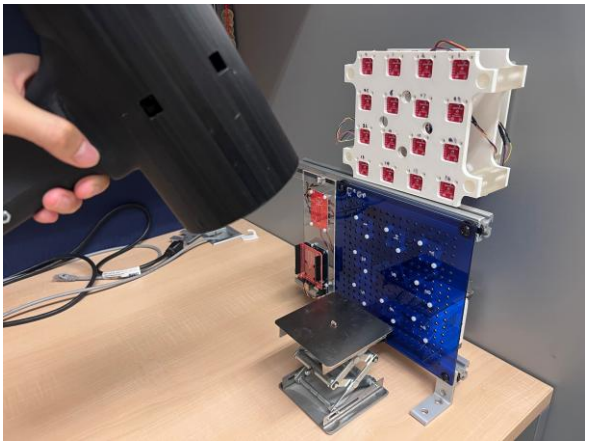


Figure 8: Ideal Alignment System Setup (the sensor array placement can be adjusted, and the hand-held device is not necessarily pointing to the internal magnet)

5.4 Real-Time Visual Assisted Alignment System

To provide intuitive guidance for the operator in aligning the external magnet with the required rotation axis, a real-time visualization system has been developed in Python. This system is integrated with the localization system and rotation axis calculator, offering a real-time display of the magnets and the rotation axis. The user interface is shown in the figure below.

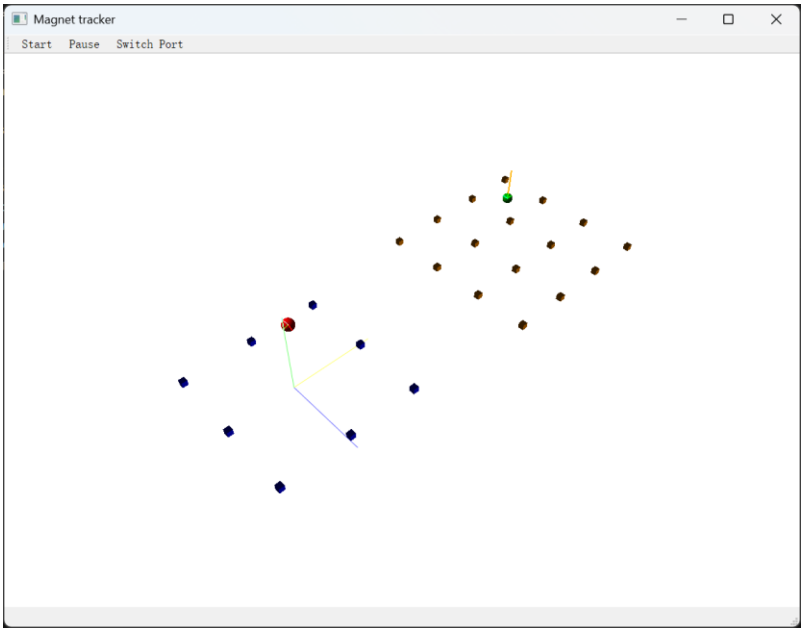


Figure 9: Real-Time Visual Assisted Alignment System Interface

The interface activates as soon as the main program is executed, displaying the defined sensor array positions in a 3D interactive window. Different sensor array placements can yield varying performances. Two sensor arrays should be connected before starting the main program to ensure proper serial port communication setup. By default, the program is in a "Paused" state. Once the sensor array calibration is completed, the user can click the "Start" button to begin receiving data from the sensor array connected to the first USB port, initiating the real-time display of the internal magnet. The system will show both the location (indicated by a red ball) and the magnetic moment direction (indicated by a red line), continuously updating the real-time location and direction (as described in alignment procedure 3). The user can click the "Pause" button to freeze the display at any time.

Two candidate rotation axes will also be displayed in green and cyan for reference. Assuming the internal magnet remains static, the user can then click the "Switch Port" button to change the USB connection and press the "Start" button again to accept data from the MLX sensor array. The external magnet and the required rotation axis will then be displayed on the screen in real time, enabling the operator to easily align the hand-held device with the required rotation axis using visual aids.

6 Future Work

Firstly, due to the malfunction of the MLX sensors (refer to Appendix), the real-time display of the required rotation axis has not been completed. However, the framework for real-time display and calculation has been established. New sensors can be installed in the sensor array, and their outputs can be integrated into the existing framework to enable the functionality mentioned above. Please refer to the README file in the program package for further details.

Secondly, a spherical angle system is employed in this project. When the polar angle is equal to 0 or π, the optimization encounters a singularity problem, meaning that changes in the azimuthal angle do not affect the magnetic moment direction. In this case, the optimizer struggles to find the optimal azimuthal angle that minimizes the loss function, as it perceives a flat loss function landscape concerning the azimuthal angle. This can lead to inaccurate results from the localization algorithm. Moving forward, Cartesian coordinates could be utilized to eliminate the singularity issue.

Thirdly, the placement of multiple sensors can be optimized in the future. Different layouts will yield varying system performances. An optimized layout can enhance the accuracy of localizing the internal magnet while preventing saturation of the sensor array from the external magnet. Additionally, data processing techniques such as the Kalman Filter can be applied to enhance data stability and safety.

Fourthly, precision requirements and error tolerance ranges should be carefully defined based on surgical specifications. Once these parameters are established, the stable operational range of the system can be determined using simulated error graphs. A systematic evaluation of the alignment system is also necessary to quantify and verify accuracy and failure rates under real working conditions.

Finally, the user interface and interaction experience can be further improved to provide more intuitive guidance. Various guiding schemes, such as audio prompts, could be added to enhance operator convenience. Existing alignment systems in the medical field can serve as examples for refining this system to meet practical requirements.

7 Conclusion

This report presents the development of a real-time visual assisted alignment system for accurately positioning an external magnet relative to an internal magnet within a surgical context. Despite challenges, such as sensor malfunctions and optimization issues, a robust framework has been established for localization and display. Future work will focus on

enhancing sensor capabilities, optimizing sensor placements, addressing singularity problems, and refining user interface interactions. Ultimately, the goal is to improve the accuracy and reliability of the alignment system, ensuring it meets the demands of practical surgical applications.

References:

[1] “Endoscopic Submucosal Dissection,” *www.hopkinsmedicine.org*, Mar. 28, 2022. https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/endoscopic-submucosal-dissection

[2] E. Yang, “Hand-Held Magnetic Field Generator for Gastrointestinal Tract Interventional Device,” MIE8888Y MEng Project, Dec. 2024.

[3] A. W. Mahoney and J. J. Abbott, “Generating Rotating Magnetic Fields With a Single Permanent Magnet for Propulsion of Untethered Magnetic Devices in a Lumen,” *IEEE Transactions on Robotics*, vol. 30, no. 2, pp. 411–420, Jan. 2014, doi: https://doi.org/10.1109/tro.2013.2289019.

[4] H. Mayer, “Summer Research Project S2025,” May 2025.

Appendix

A Malfunction of MLX90393 sensor

The Malfunction is determined by using both MLX90393 and MMC5983MA to measure the earth field. Since the earth field will not change in the dimension of sensor array, each sensor's measurement in the sensor array when no magnets around can be viewed as a sample of the earth field and they should be the similar.

MMC5983MA (micro-Tesla)	x	y	z
S1	11.39	-22.03	-37.39
S2	12.62	-8.13	-34.32
S3	10.5	-16.71	-44.68
S4	10.85	-7.71	-33.26
S5	10.01	-15.74	-41.56
S6	10.03	-18.35	-45.72
S7	9.58	-7.48	-38.17
S8	10.19	-15.45	-42.07
Average	10.64625	-13.95	-39.64625

MLX90393 (micro-Tesla)	x	y	z
S1	-18.64	-93.72	-56.74
S2	-11.3	-92.96	-58.27
S3	-26.88	3.46	-73.23
S4	-6.53	-46.9	-56.35
S5	-18.46	-45.26	-75.36
S6	-6.75	-45.5	-57.63
S7	-22.74	-45.95	-55.36
S8	-3.14	-94.6	-55.4
S9	11.17	-61.09	-55.53
S10	-0.98	-95.46	-75.05
S11	-10.78	-71.07	-55.6
S12	6.32	-70.92	-56.43
S13	-14.41	-59.17	-56.19
S14	-0.58	-58.14	-54.1
S15	-15.51	-83.19	-65.99
S16	-3.95	-94.71	-55.6
Average	-8.9475	-65.9488	-60.1769

Multiple tests have been conducted. In each test, the two sensor arrays were placed at the same location, and the sensor coordinates are aligned. 300 samples were taken for every sensor and the average value of all the sensors in x, y, and z axes are calculated. By comparing the average value, it is clear that the MLX90393 fails to measure the earth field correctly. This mainly comes from the great differences between MLX90393 sensor measurements. MMC5983MA sensor measurements also have some discrepancies but the MLX90393 is more severe, indicating a possible sensor malfunction.

B Comparison between Cosine Distance and Euclidean Distance

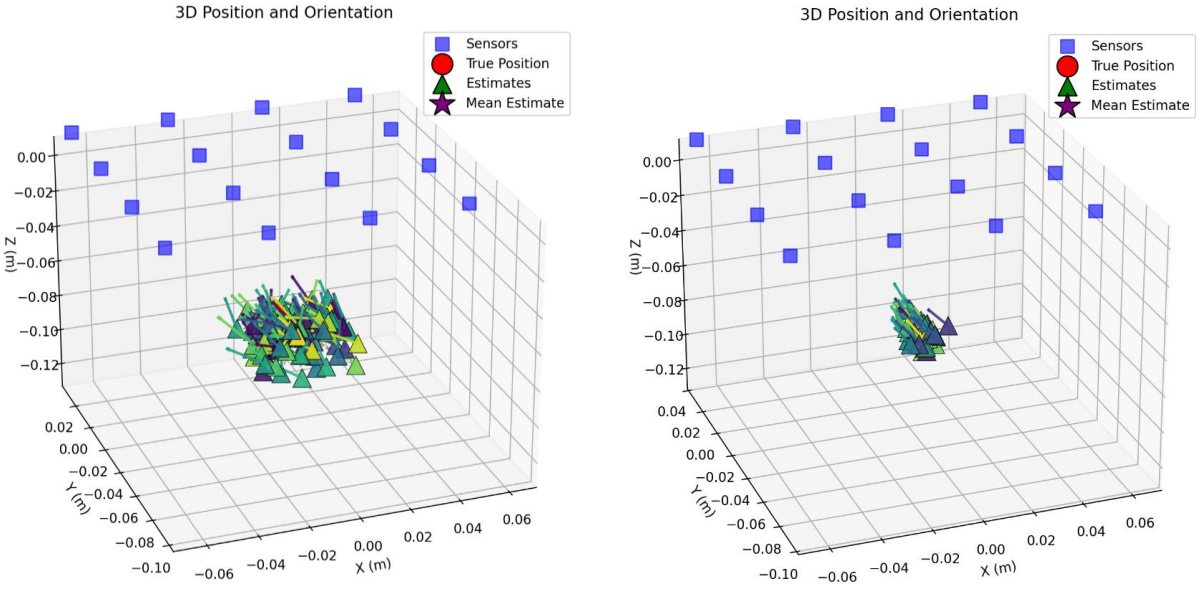


Figure 10: Internal magnet, Ture location: (0, 0, -0.1), True orientation: along positive y-axis, 100 optimizations